Plantation Management Research Cooperative

Warnell School of Forestry & Natural Resources

UNIVERSITY OF GEORGIA

2026 PMRC Annual Advisory Meeting

Clustering growth and yield across species.

Quentin Boccaleri[1], Bronson Bullock[1,], Stephen Kinane[1], Sheng-I Yang[1], Pete Bettinger[1], and Bruno da Silva[1]

[1] *D.B. Warnell School of Forestry and Natural Resources, University of Georgia, 180 East Green Street, Athens, Georgia 30602.*

1. Introduction

- **Natural multi-cohort and multi-species stands** (referred to hereafter as "complex stands") **have limited representation in the literature.**
- **Complex stands make up nearly 200 million acres of forest land in the southeastern United States.**
- **Modeling on the functional group level can lead to a more efficient use of limited data.**
- This project will **examine multiple species growth curves and aggregate them to them into functional groups.**
- AOI: **Southeastern Plain of Alabama.**

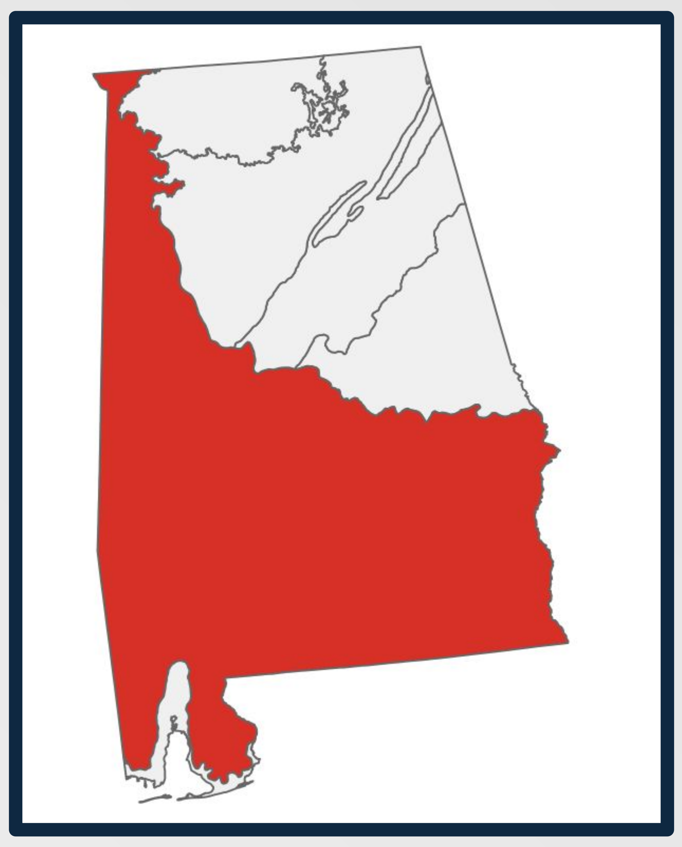
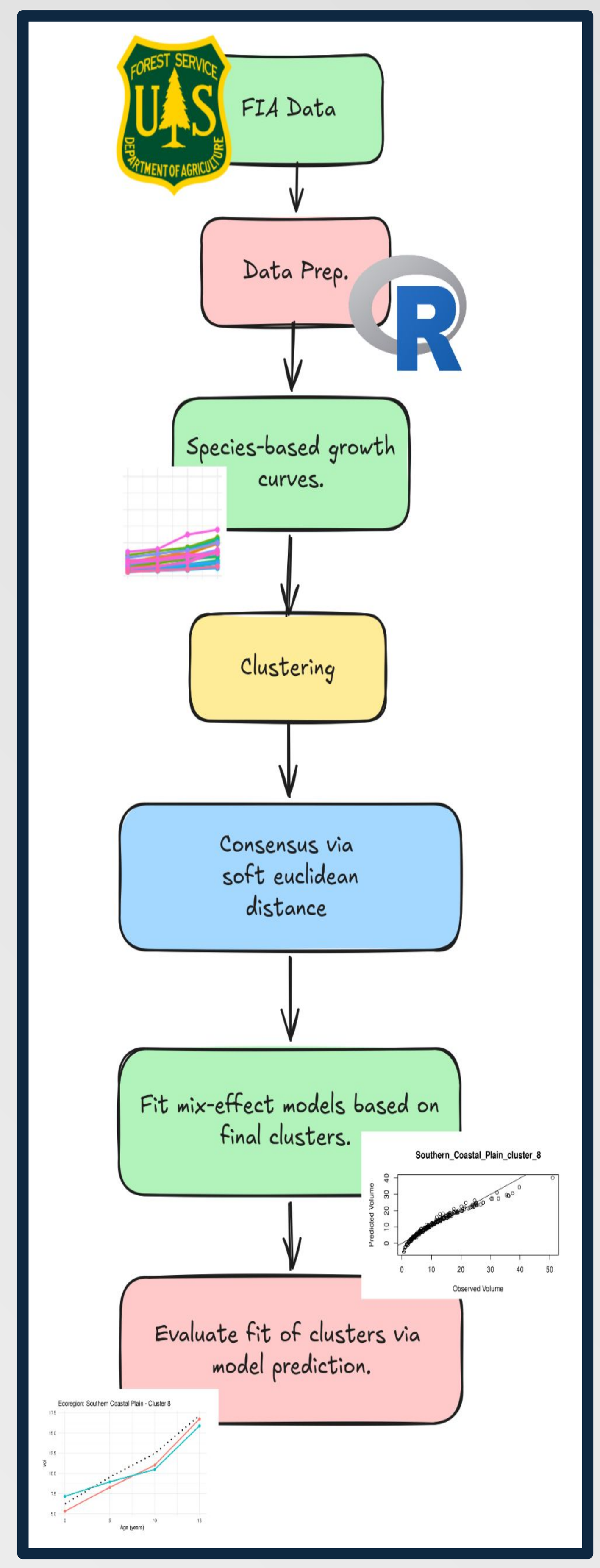


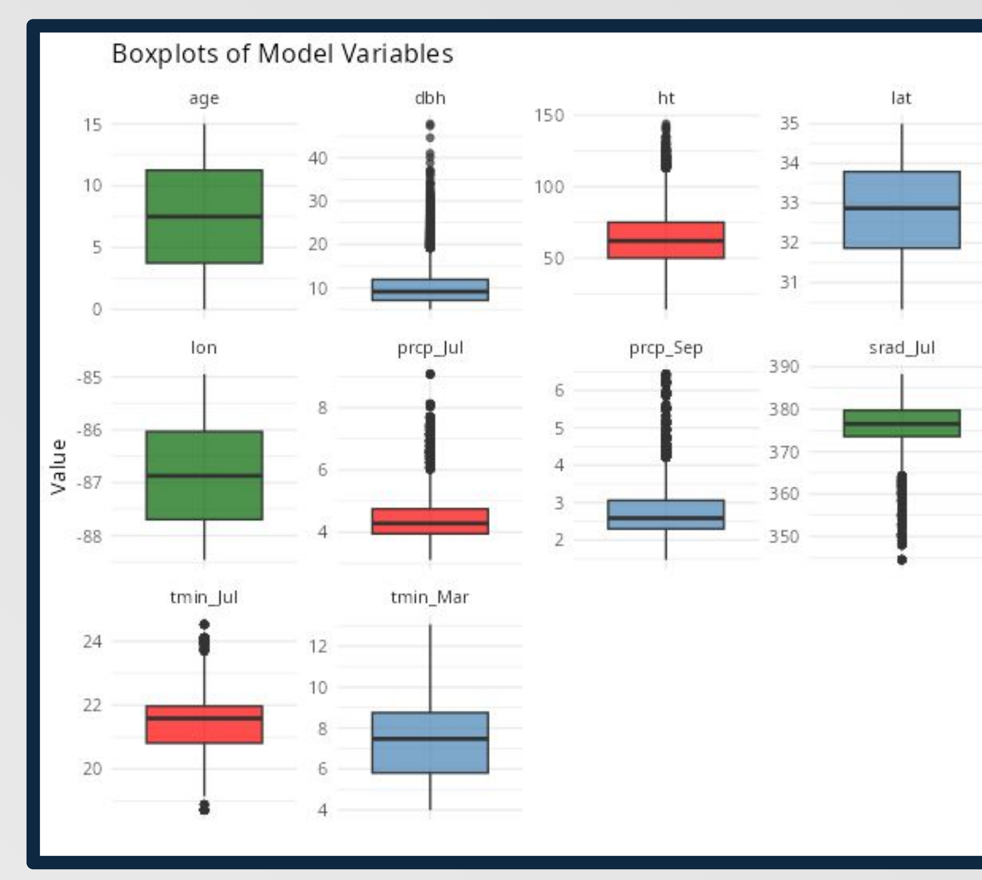
Figure 1 - The state of Alabama with the area of interest highlighted in red.

2. Material and Methods



- Data via the **FIA datamart and DAYMET.**
- Data was filtered for trees that have 15 **years of recorded growth.**
- **Species growth curves were built using median net volume (cubic ft.).**
- **8 clustering methods were used:** k means, hierarchical, DBSCAN, HDBSCAN, spectral, affinity, BIRCH, and fuzzy.
- **Final clusters assigned via a consensus analysis** using soft euclidean distance.
- **Mixed effect models were fit for the final clusters.**

```
Net Volume(cuft) = Plot Coordinates + HT + DBH +
Climate Parameters + (1 | tree_id)
```



3. Results - Southeastern Plain

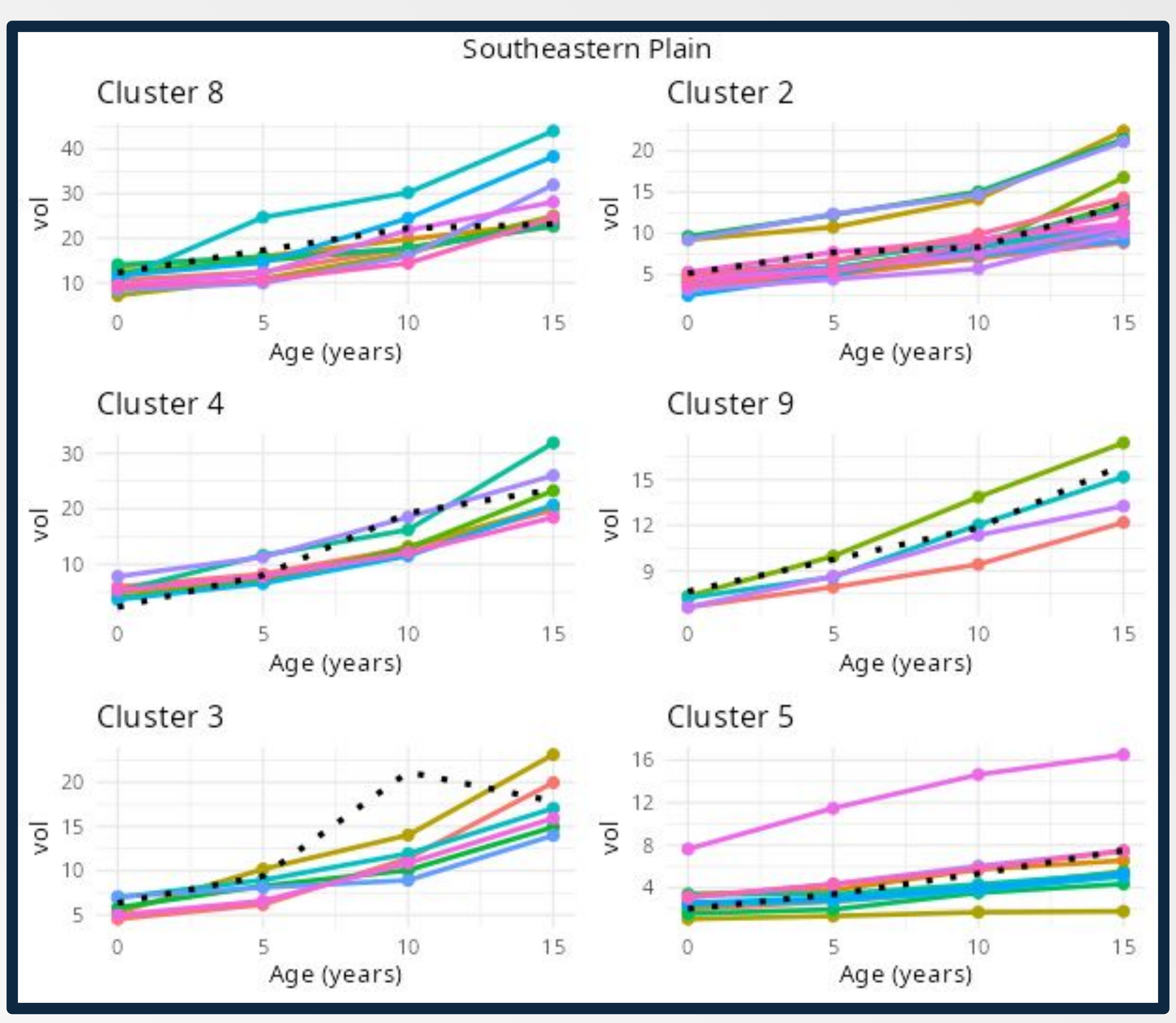


Figure 2 - Each figure shows the growth curves for each cluster and the model prediction (dotted).

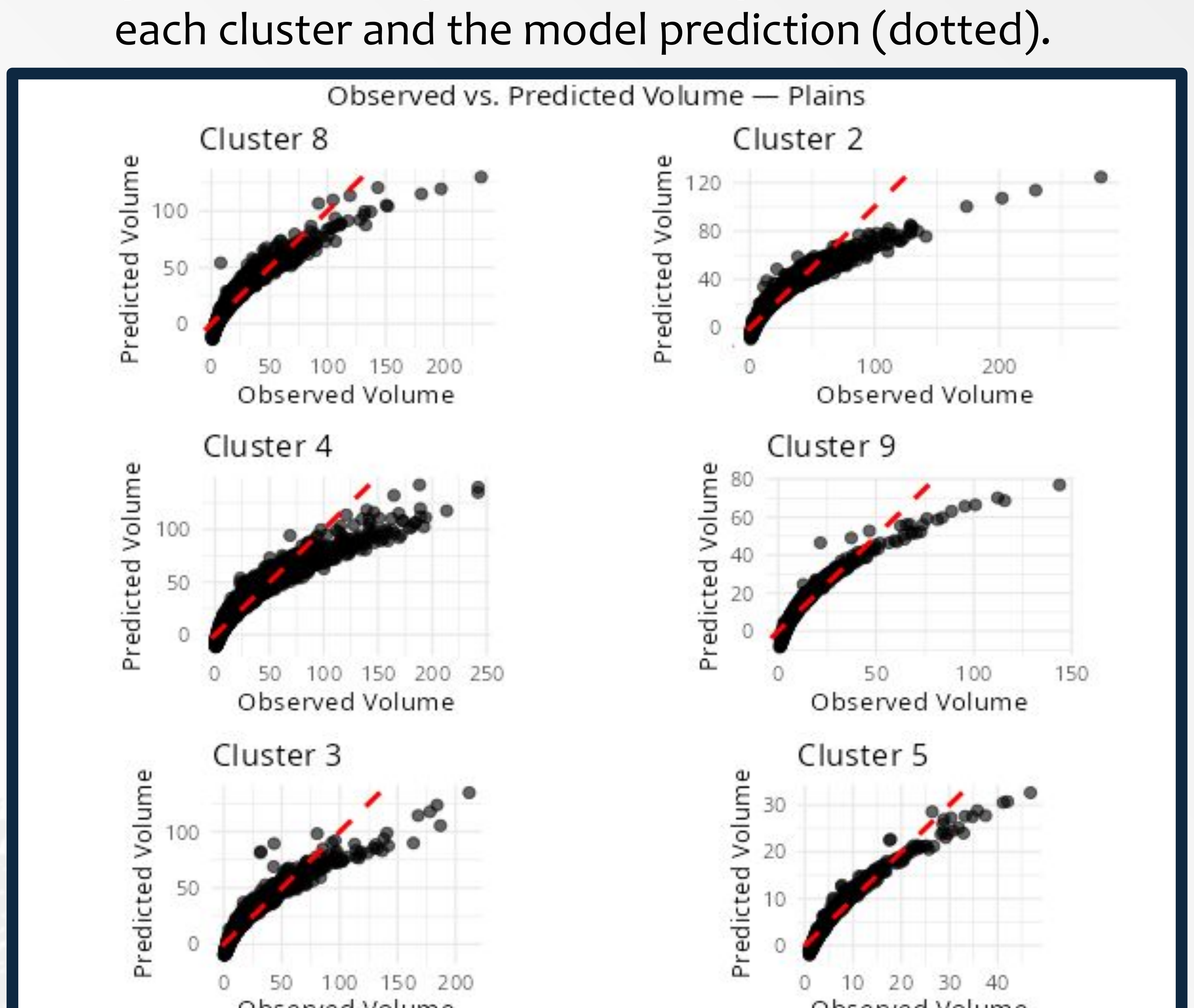


Figure 3 - Observed vs. predicted plots for the mix-effects model for each cluster.

4. Future Work

- **All clusters over predicted volume as observations got larger - indicating a nonlinear equation may be needed.**
- **There are signs that these models will have performance issues on validation data.**
- More work is needed to **optimize model parameter selection and equation from for each cluster.**

5. Acknowledgements

Special thanks to Dr. Pete Bettinger and the eYield team for providing funding and support for this project in partnership with the PMRC. This research is funded by USDA NIFA grant 2024-67023-43581.

eYield Website

TEMP DO NOT SCAN

R Shiny App